

Probability Distributions

STAT 4530/6530
Statistical Inference for Data Scientists

Probability

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Defining probability in terms of a long-run sequence of like observations is not always appropriate.

- For example, it is not meaningful for the probability that intelligent life exists elsewhere in the universe or for the probability that a new business is successful, because no long-run sequence of observations is available.
- In such cases, we must then rely on subjective beliefs rather than objective data to characterize the probability.

In fact, an alternative definition of probability is the **subjective definition of probability**: the probability of an outcome is the degree of belief that the outcome will occur, based on all available information.

Sample Spaces and Events

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Example: you ask three people whether marijuana use should be legal, with possible responses *yes* and *no*

- Sample space: $S = \{(\text{yes}, \text{yes}, \text{yes}), (\text{yes}, \text{yes}, \text{no}), (\text{yes}, \text{no}, \text{yes}), (\text{no}, \text{yes}, \text{yes}), (\text{yes}, \text{no}, \text{no}), (\text{no}, \text{yes}, \text{no}), (\text{no}, \text{no}, \text{yes}), (\text{no}, \text{no}, \text{no})\}$
- Consider events $A = \{(\text{yes}, \text{yes}, \text{yes}), (\text{no}, \text{no}, \text{no})\}$ and $B = \{(\text{yes}, \text{yes}, \text{yes}), (\text{yes}, \text{yes}, \text{no}), (\text{yes}, \text{no}, \text{yes}), (\text{yes}, \text{no}, \text{no})\}$

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- The **union** of events A and B , denoted by $A \cup B$, consists of the elements of the sample space that are in A or B

Mathematical Formulation of Probability

Let $P(A)$ denote the probability of an event A . All probabilities must satisfy the following axioms:

- $P(A) \geq 0$ (probabilities are nonnegative)
- For the sample space S , $P(S) = 1$ (the total probability for all possible outcomes equals 1)
- If A and B are disjoint (meaning that their intersection is empty), then $P(A \cup B) = P(A) + P(B)$

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Other probability rules follow from these axioms:

- $P(A^c) = 1 - P(A)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Note: it is common to denote $P(A \cap B)$ by $P(A, B)$, so the last identity can also be expressed as

- $P(A \cup B) = P(A) + P(B) - P(A, B)$

Conditional Probability

The probability of an event A , given that an event B occurred, is called the **conditional probability** of A given B , and is denoted by $P(A|B)$.

We have

$$P(A|B) = \frac{P(A, B)}{P(B)}.$$

Placing $P(B)$ in the denominator amounts to restricting consideration to the sample elements in B :

$$P(A|B) = \frac{\text{Diagram 1}}{\text{Diagram 2}} = \frac{P(A \text{ and } B)}{P(B)}$$

The probability in the denominator is always for the “given” event

Example: Diagnostics for Disease Screening

A mammogram is an imaging device for diagnosing breast cancer.

- A diagnostic test is *positive* (+) if it indicates that the disease is present and *negative* (−) if it indicates that the disease is absent
- We denote the true disease status by $D = \text{yes}$ and $D^c = \text{no}$
- The probability that the diagnostic test is positive given that the person has the disease, $P(+|D)$, is called the **sensitivity**
- The probability that the diagnostic test is negative given that the person does not have the disease, $P(-|D^c)$, is called the **specificity**

Example: Diagnostics for Disease Screening

Disease Status	Mammogram Diagnosis		Total
	Positive (+)	Negative (−)	
Yes (D)	0.86	0.14	1.0
No (D^c)	0.12	0.88	1.0

Let us suppose that the probability that a woman has the disease, $P(D)$, is 0.01. Using the information in the table above, compute

- The probability of having the disease given a positive diagnosis
- The probability of not having the disease given a negative diagnosis

Independent Events

Two events A and B are **independent** if

$$P(A, B) = P(A)P(B).$$

This condition is equivalent to the following conditions:

- $P(A|B) = P(A)$
- $P(B|A) = P(B)$

Random Variables

We are generally interested in computing probabilities of numerical summaries of the sample space elements, known as random variables.

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We can, for instance, define a random variable X to represent the number of people who say *yes*.

- What values does the random variable X assign to the elements of S ?

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Note: random variables are typically represented by upper-case letters, and specific values they can assume are represented by lower-case letters

- For example, $x = 2$ is a possible value for the random variable X

Probability Distributions

A **probability distribution** is a function that characterizes the probabilities of possible outcomes for a random variable.

- The nature of the function depends on whether the random variable is discrete or continuous.

A random variable is **discrete** if the set of possible outcomes is finite or, in the case that it is infinite, can be put into a one-to-one correspondence with the set of nonnegative integers.

- In the latter case, this intuitively means that there are gaps in between the values of the possible outcomes.

On the other hand, a random variable is **continuous** if the set of possible outcomes is an uncountably infinite set.

- We are often interested in random variables that can take on a continuum of values without any gaps.

Probability Distributions

A discrete random variable X has a **probability mass function (pmf)** $f(x)$, which gives the probability that the random variable is exactly equal to a given value x :

$$f(x) = P(X = x).$$

The probability mass function satisfies the following conditions:

$$f(x) \geq 0 \text{ for each possible outcome } x$$

and

$$\sum_{\text{all possible } x} f(x) = 1.$$

Probability Distributions

Let's go back to the example of three people being asked whether marijuana use should be legal, with the random variable X representing the number of people who say yes.

If we assume that the probability that a person responds yes is equal to p , then X follows a **binomial** distribution with $n = 3$ “trials” (denoted by $\text{Bin}(n, p)$)

The probability mass function is

$$f(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & \text{for } x = 0, 1, \dots, n \\ 0 & \text{for all other } x \end{cases}$$

where

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}.$$

Probability Distributions

A continuous random variable X has a **probability density function (pdf)** $f(x)$, which is used to assign probabilities to sets of possible outcomes (typically intervals of real numbers).

Suppose X is real-valued. Then $f(x)$ has the following properties:

- $f(x) \geq 0$ for all real numbers x
- $f(x)$ is piecewise continuous
- $\int_{-\infty}^{\infty} f(x)dx = 1$

If a and b are real numbers with $a < b$, then

$$P(a < X < b) = \int_a^b f(x)dx.$$

Probability Distributions

An example of a continuous random variable is a uniform distribution on the interval $[a, b]$, denoted by $U[a, b]$.

- All intervals of the same length contained in $[a, b]$ have the same probability

The probability density function is

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b] \\ 0 & \text{for all other } x \end{cases}$$

- Can you draw a plot of the pdf of a $U[0, 1]$ distribution?

Probability Distributions

For both discrete and continuous random variables, the probability distribution is also characterized by the cumulative distribution function.

The **cumulative distribution function (cdf)** $F(x)$ of a random variable X indicates the probability that X is less than or equal to any real number:

$$F(x) = P(X \leq x) \text{ for all real numbers } x.$$

- Can you compute the cdf of a $U[0, 1]$ distribution?

Expectations of Random Variables

A probability distribution can be characterized by certain descriptive measures.

- One such measure is the mean (also called the expected value), which represents the mean of the values that the corresponding random variable would assume in an indefinitely long sequence of independent observations from the distribution.

If X is a discrete random variable with pmf $f(x)$, then its **expected value** is

$$E(X) = \sum_{\text{all } x} xf(x),$$

and if X is a continuous random variable with pdf $f(x)$ defined on the real number line, then its **expected value** is

$$E(X) = \int_{-\infty}^{\infty} xf(x)dx.$$

- For $X \sim U[0, 1]$ (meaning for a random variable X that follows a $U[0, 1]$ distribution), compute $E(X)$.
- For $X \sim \text{Bin}(3, 0.5)$, compute $E(X)$.

Standard Deviation and Variance

A probability distribution can also be characterized in terms of its spread, which reflects the long run variability of observations from that probability distribution.

Consider a random variable X with mean $E(X) = \mu$. The **variance** $Var(X)$ is defined as:

$$Var(X) = E[(X - \mu)^2],$$

and the standard deviation $SD(X)$ is the positive square root of the variance.

- Compute the variance of a random variable that follows a $U[a, b]$ distribution.
- Compute the variance of a random variable that follows a $\text{Bin}(3, 0.5)$ distribution.

Moments

Expected values such as $E(X)$ and $E(X^2)$ are examples of **moments** of a random variable X . If k is a nonnegative integer, the k -th moment is defined as $E(X^k)$.

If $E(X) = \mu$, we define the k -th **central moment** as $E[(X - \mu)^k]$.

The third central moment $E[(X - \mu)^3]$ describes the **skewness** of the distribution:

- It is equal to 0 when the distribution is symmetric
- It is positive when the distribution is skewed to the right (has a longer “right tail”)
- It is negative when the distribution is skewed to the left (has a longer “left tail”)

Let σ denote $SD(X)$. To summarize the skewness in a way that does not depend on the units of measurement, we define

$$\text{Skewness Coefficient} = \frac{E[(X - \mu)^3]}{\sigma^3}.$$

Expectations of Linear Functions of Random Variables

Consider a random variable X and real numbers a and b .

Suppose that we define $Y = a + bX$.

- Compute $E(Y)$
- Compute $Var(Y)$

Let μ denote $E(X)$ and σ denote $SD(X)$. It is common to **standardize** the random variable X as follows so that it has mean 0 and standard deviation 1:

$$Z = \frac{X - \mu}{\sigma}.$$

Z represents the number of standard deviations that X falls from its mean. For example, when $Z = -0.8$, then X falls 8/10 of a standard deviation below μ .

Quantiles of a Probability Distribution

For a continuous random variable, the p -th quantile (equivalent to the $100 \cdot p$ percentile) is the point q at which $F(q) = p$.

- For example, the 0.50 quantile (the **median**) is the point q at which $F(q) = 0.50$.
- The point q is determined by applying the inverse of the cdf to p : $q = F^{-1}(p)$.

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It can be shown that if X is a continuous random variable with cdf $F(\cdot)$, then the random variable $Y = F(X)$ has a $U[0, 1]$ distribution.

- This fact is useful for simulating from distributions.
- To generate a random variable X having cdf $F(\cdot)$, generate $Y \sim U[0, 1]$ (which any statistical software can do), and then set $X = F^{-1}(Y)$.

Joint and Marginal Probability Distributions

A **joint probability distribution** specifies probabilities for all possible combinations of values of multiple random variables.

Consider the bivariate case with random variables X and Y .

- For continuous random variables, the joint probability distribution is specified by a joint pdf $f_{X,Y}(x,y)$
- For discrete random variables, the joint probability distribution is specified by a joint pmf $f_{X,Y}(x,y) = P(X = x, Y = y)$

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We can compute the **marginal distribution** of any single random variable (that is, the distribution of the variable considered by itself without reference to any other random variable) from the joint distribution by integrating or summing over the other variables. Suppose, for instance, we want the marginal distribution for X :

- If the variables are continuous, $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)dy$
- If the variables are discrete, $f_X(x) = \sum_{\text{all } y} f_{X,Y}(x,y)$

Example

Family Income	Happiness			Total
	Not too happy	Pretty happy	Very happy	
Below average	0.080	0.198	0.0079	0.357
Average	0.043	0.254	0.143	0.440
Above average	0.017	0.105	0.081	0.203
Total	0.140	0.557	0.303	1.00

This table presents results from a General Social Survey in which respondents were asked, “Would you say that you are very happy, pretty happy, or not too happy?” and “Compared with American families in general, would you say that your family income is below average, average, or above average?”

- Let X denote family income and Y denote happiness. What are the marginal distributions?

Conditional Probability Distributions

A **conditional probability distribution** specifies the probabilities for the outcome of one random variable, conditional on the outcome for another random variable.

When X and Y have joint pmf or pdf $f_{X,Y}(x,y)$ and marginal pmfs or pdfs $f_X(x)$ and $f_Y(y)$, the conditional probability function for Y given $X = x$ is:

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}.$$

- For the General Social Survey example (see previous slide), compute the conditional probability for Y given $X = \text{average}$.

Conditional Expectations

We can characterize a conditional distribution by moments (just as with other probability distributions). For example, for continuous random variables X and Y , the **conditional expectation** of Y given $X = x$ is

$$E(Y|X = x) = \int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy.$$

A useful result related to conditional expectations is the **law of iterated expectation**:

$$E(Y) = E[E(Y|X)],$$

where the outer expectation on the right side is with respect to the distribution of X .

Expectations of Sums of Random Variables

A useful property for jointly distributed random variables $Y_1, \dots, Y_n$ is that

$$E(Y_1 + Y_2 + \dots + Y_n) = E(Y_1) + E(Y_2) + \dots + E(Y_n).$$

- Prove this property for the bivariate case with random variables X and Y .

Independence of Random Variables

Random variables X and Y with marginal probability functions $f_X(x)$ and $f_Y(y)$ are **independent** when

$$f_{X|Y}(x|y) = f_X(x) \text{ and } f_{Y|X}(y|x) = f_Y(y)$$

for all possible values x of X and y of Y . In this case, the joint probability function factors as:

$$f_{X,Y}(x,y) = f_X(x)f_Y(y) \text{ for all } x \text{ and } y.$$

We say that random variables $Y_1, \dots, Y_n$ are **mutually independent** if their joint probability function factors as a product of the marginal probability functions:

$$f_{Y_1, \dots, Y_n}(y_1, \dots, y_n) = f_{Y_1}(y_1) * \dots * f_{Y_n}(y_n) \text{ for all values } (y_1, \dots, y_n).$$

Covariance and Correlation

The **covariance** between random variables X and Y with $E(X) = \mu_X$ and $E(Y) = \mu_Y$ is

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)].$$

The covariance measures the association between the random variables. It can take on any real number as a value, and its size depends on the units of measurements. To make it free of the units of measurement, we can standardize the differences $X - \mu_X$ and $Y - \mu_Y$ by the corresponding standard deviations σ_X and σ_Y . This yields the **correlation** between X and Y :

$$\text{Corr}(X, Y) = E \left[\left(\frac{X - \mu_X}{\sigma_X} \right) \left(\frac{Y - \mu_Y}{\sigma_Y} \right) \right] = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

Note: If two random variables are independent, their correlation will be zero. However, the converse does not hold.

Reference

Agresti, A., & Kateri, M. (2021). Foundations of Statistics for Data Scientists: With R and Python (1st ed.). Chapman and Hall/CRC.